

JUAN MIGUEL CERDA ORDAZ

Game Development Engineer | Software Engineer | Technical Lead | Co-Director

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🌐 mickordaz.github.io/Portfolio

Summary

Game Development Engineer and Software Engineer with 7+ years spanning indie development, engineering leadership, 3D simulation systems, and AI-driven research platforms. Built a custom C++ 3D graphics engine with a live Nvidia Omniverse connector during the platform's early development — featured at Nvidia GTC 2021. Co-directed and shipped award-winning games, developed cell behavior and cancer research bio-simulations, architected multi-agent AI swarms, and delivered clean optimized code across custom C++ renderers and modern web stacks.

Education

Universidad de Artes Digitales

B.S. in Game Development Engineering

2018 – 2021

Guadalajara, Mexico

Experience

Founder, Developer & Tech Lead

2019 – 2026

Critical Slash — Independent Studio

Mexico

- Founded studio and solo-developed Heaven Crawler over 6 years from concept to a fully playable experience; partnered with Carbon Machina Studio in 2024 to scale production for commercial release.

Lead Programmer & Co-Director

2021 – 2026

Carbon Machina Studio

Mexico

- Led multidisciplinary engineering team in Unity and Unreal Engine: core gameplay, advanced AI behavior, gamefeel, UI, and editor tooling.
- Co-directed creative and technical vision as IP owner — characters, story, and gameplay direction — while leading all engineering. Showcased at Gamescom & Latin American Game Showcase; won Grand Prize at Supernova Game Devs Challenge 2025 (unanimous).

Professor of Video Game Prototyping

2022 – 2023

Universidad de Artes Digitales

Guadalajara, Mexico

- Instructed university students on rapid prototyping, core gameplay loops, production mechanics, and clean code architecture in Unity.

Medical AI & Research Engineer

2022 – 2026

Bowhead Health & Good Place Games OÜ

Remote

- Engineered a multi-agent AI swarm system for biomedical research with a real-time 3D Virtual Lab (React / Python / FastAPI); developed bio-simulations modeling cell behavior, growth patterns, and cancer progression for research analysis.
- Built XR apps for Apple Vision Pro (Unity / visionOS); contributed API connectors to Minicule, a drug discovery platform (PubMed, OpenAlex, USPTO).

Projects

Heaven Crawler | *Unity, C#, Behavior Trees, Procedural Gen, IK* | Lead Programmer & Co-Director

2019 – 2026

- Action-adventure roguelite dungeon crawler. Solo-developed for 6 years, then co-led full production team. Grand Prize Winner, Supernova Devs Challenge 2025 (unanimous).
- Most Anticipated Game Nominee (Game Effect Awards, 2025); Official Indie Showcase Selection (Mexican Entertainment System, 2025–2026); showcased at Gamescom & Latin American Game Showcase.

Biogame AI | *React, Three.js, Python, FastAPI, LangGraph, Neo4j, ChromaDB, Docker* | Sole Architect

2024 – 2025

- SaaS multi-agent AI bioinformatics platform with an isometric 3D lab. Built end-to-end: LangGraph orchestration, 12+ tool modules, knowledge graph pipeline, Stripe billing, deployed on Google Cloud Run.

Vivarium Engine | *C++, OpenGL, DirectX, Nvidia Omniverse API, PhysX* | Solo Developer

2020 – 2021

- Custom C++ 3D game engine with deferred PBR rendering, PhysX, and VR support; live multi-client scene sync via Nvidia Omniverse. Featured Developer at Nvidia GTC 2021; featured on Nvidia developer blog.

Docs Slayer | *React 19, TypeScript, Hono, Cloudflare Workers, Gemini AI, Firebase* | Sole Founder

2025 – 2026

- AI documentation engine for game studios: GitHub-connected, Gemini-powered deep code scan, human diff review workflow. Built solo end-to-end. Closed beta.

Other Projects: Ramen Abyss, Mecha K.O., Doppelbio, Minicule, Propermax, Exact Agent, Mental Chat AI, Chrimp Desktop Pet, American Trail, Cyberdogz, SOLBAG, Slug Slayer, Skate & Slash

Technical Skills

Languages: C#, C++, Python, JavaScript, TypeScript, HTML5, CSS3

Engines & Tools: Unity, Unreal Engine, React, Three.js, FastAPI, Node.js, Docker, Git, Custom C++ Engines

Specialties: Core Gameplay Engineering, Gamefeel & Polish, Systems Architecture, AI Behavior & Agent Swarms